

Urchin grazing experiment and faecal production

1. Set up as many flow-through tanks as can be feasibly sampled (I had 30 for urchins and autogenic controls), ideally with lighting on a 12-12-h light-dark cycle (I set this up using LED strips and a timer switch)
2. Collect at least 5 times as many *Strongylocentrotus droebachiensis* individuals as the number of urchin tanks (I had 15 urchin tanks, so 75 urchins)
3. Also collect enough *Laminaria hyperborea* laminae to have at least one per 25 urchins (cut above meristem, i.e. stipe-lamina interface, to allow regrowth)
4. Pick the most viable urchins and haphazardly spread across tanks, placing 5 in each
5. Cut *L. hyperborea* laminae into frond segments, avoiding basal and distal sections, and place one segment in each tank
6. Feed *S. droebachiensis* ad libitum on *L. hyperborea* for at least 2 days to purge gut while maintaining a gut full of the desired material (the initial frond should be enough but add more if needed)
7. After at least 2 days, collect, as in step 3, enough *L. hyperborea* laminae to have at least one per 5 urchins and cut as in step 5
8. This time the frond portions will be larger, can be made up of several fronds and should be roughly of equal mass (I had ~200 g wet mass per tank, but judging by the grazing rates I measured, ~100 g should be more than enough)
9. Make sure to not mix tissue from multiple sporophytes in one tank, but instead split tissue from one sporophyte across two tanks, one grazing and one control tank
10. Weigh two frond portions per sporophyte to 1-mg accuracy (I tried buoyant mass initially but 200 g of kelp plus beaker with water maxed out the balance, so I had to go with briefly dripped wet mass; in any case, do not blot the kelp because you will remove the microbiome)
11. Shortly before adding the weighed kelp to the tanks, clean up the accumulated faeces in each urchin tank using the 1-mm net and discard (this is best done by transferring the urchins to another tank)
12. As you are replacing the urchins into their clean tank, measure the test diameter (digital calipers) and dripped mass (same balance, 1-mg accuracy) of each
13. Start the experiment by adding the kelp portions from each sporophyte to an urchin and a control tank, noting the date and time, sporophyte identity, initial lamina mass and initial urchin masses and test sizes for each tank (see below for example data)
14. Deploy temperature loggers, recording at a measurement interval of 30 min, in a random sample of the experimental tanks
15. Leave urchins to graze on kelp for between one and two weeks
16. Terminate the grazing experiment by first taking out the remaining lamina and weighing as in step 10, noting the date and time, sporophyte identity and final lamina mass
17. Excise two discs (e.g. 5-cm diameter) from this lamina, transfer to a ziplock bag for the mesograzer grazing experiment and note down the residual lamina mass (only for urchin tanks)
18. Next, take out the urchins one by one, take their measurements as in step 12, noting final urchin masses and test sizes, and transfer to another tank
19. Finally, with only the faeces left in the tank, collect them using the 1-mm net, drain briefly and add to a tray with seawater to measure and note down buoyant mass to 1-mg accuracy (the easiest way to do this is to gently shake the net containing all faeces from side to side until they form a ball, then invert the net into the tray on the balance and use the water in the tray to wash out remaining faeces clinging to the net)
20. Transfer ~2 g buoyant mass of faeces to a test tube or ziplock bag for the mesograzer grazing experiment and note down the residual faecal mass
21. Transfer the remaining lamina to one labelled ziplock bag and remaining faeces to another

22. Place ziplock bags in freeze-dryer, making sure that they are partly open, and dry for 48 hours
23. Measure dry mass of the remaining lamina and faeces to 0.1-mg accuracy
24. Crumple up and homogenise the lamina and faeces in their respective ziplock bags
25. Transfer a sample of each homogenate to a labelled screw-top microtube containing zirconium beads and mill to a fine powder
26. Send cardboard cryoboxes filled with these tubes to me in the UK between August and December for biochemical analyses (if there is lack of space, prioritise faecal samples)
27. Mill the remaining homogenate into a powder for the agar-based mesograzers assay

Mesograzers grazing experiment

1. Set up as many grazing arenas as there were urchin tanks in one of the controlled temperature rooms, setting it to a sensible average temperature between 10 and 15°C (my urchin experiment ran at 13–16°C)
2. Collect amphipods and/or isopods and feed ad libitum on *L. hyperborea* or starve (if mesograzers faeces are expected to interfere) in holding tanks
3. Transfer a single mesograzers to each grazing arena, taking its photograph next to a ruler or under the stereo microscope for later measurement with ImageJ and measuring its buoyant mass to 1-mg accuracy
4. Measure buoyant mass of the lamina discs and faeces from the same sporophyte to 1-mg accuracy
5. Start the experiment by placing lamina and faeces either in the same arena (choice experiment) or in separate arenas (no-choice experiment), remembering autogenic controls for each sporophyte
6. Leave mesograzers to graze for up to one week
7. Terminate the experiment by first picking out the remaining lamina and measuring buoyant mass to 1-mg accuracy and placing it in a ziplock bag
8. Next, catch the mesograzers and repeat step 3
9. Finally, let the faecal remains settle, gently siphon off most of the surface water using a 50-mL syringe and either pass the remaining arena contents through fine, pre-weighed (0.1-mg accuracy) filter paper and place the filter in a ziplock bag, or transfer to a pre-weighed (0.1-mg accuracy) test tube and centrifuge, discarding the supernatant
10. Freeze-dry the lamina and faecal remains, remembering that the ziplock bag needs to be partly open and/or the lid of the test tube partly unscrewed
11. Measure dry mass of the remaining lamina and faeces (together with filter or tube) to 0.1-mg accuracy
12. Subtract the mass of the filter or tube from the faecal dry mass
13. Unlike the urchin grazing experiment, the mesograzers experiment evidently has a discrepancy in the initial and final mass units; to overcome this, freeze-dry several faecal samples between 0.1 and 5 g buoyant mass and measure dry mass to get a conversion curve (see below for example data)
14. For the agar-based feeding assay, make up agar enriched with lamina or faecal powder and proceed as above, measuring area rather than mass grazed (this is up to you)

Example data

Lamina mass change and faecal production

Tank	Sporophyte	Deployment	Retrieval	Initial	Final	Lamina.wet	Lamina.dry	Faeces	Faeces.wet	Faeces.dry
U1	1	14.09.23 21:00	18.09.23 16:00	208.376	190.815	189.793	41.1185	30.189	30.189	1.9873
C1	1	14.09.23 21:00	18.09.23 16:00	201.103	201.899	200.306	48.3925	NA	NA	NA
U2	2	14.09.23 21:00	18.09.23 16:00	207.506	195.835	193.934	54.0186	17.551	17.551	1.255
C2	2	14.09.23 21:00	18.09.23 16:00	199.86	198.425	197.489	63.6138	NA	NA	NA
U3	3	14.09.23 21:00	18.09.23 16:00	206.231	194.185	193.954	49.625	11.682	11.682	0.6884
C3	3	14.09.23 21:00	18.09.23 16:00	208.567	208.336	207.574	45.2243	NA	NA	NA
U4	4	14.09.23 21:00	20.09.23 15:00	206.038	179.439	178.156	45.8914	33.323	33.323	2.4122
C4	4	14.09.23 21:00	20.09.23 15:00	202.185	190.797	190.102	55.201	NA	NA	NA
U5	5	14.09.23 21:00	20.09.23 15:00	199.905	176.522	175.581	46.3269	21.351	21.351	1.6046
C5	5	14.09.23 21:00	20.09.23 15:00	196.933	194.307	193.163	55.924	NA	NA	NA
U6	6	14.09.23 21:00	20.09.23 15:00	195.471	167.385	165.112	31.8731	29.933	29.933	2.0452
C6	6	14.09.23 21:00	20.09.23 15:00	197.237	190.77	189.753	44.3855	NA	NA	NA
U7	7	14.09.23 21:00	22.09.23 14:00	199.086	170.225	166.351	40.1316	43.284	43.284	3.4297
C7	7	14.09.23 21:00	22.09.23 14:00	202.573	207.591	206.509	39.4051	NA	NA	NA
U8	8	14.09.23 21:00	22.09.23 14:00	195.433	153.555	152.248	37.1201	42.188	42.188	3.3154
C8	8	14.09.23 21:00	22.09.23 14:00	194.994	200.584	199.614	46.1458	NA	NA	NA
U9	9	14.09.23 21:00	22.09.23 14:00	203.59	184.85	181.035	53.2342	38.779	38.779	2.9969
C9	9	14.09.23 21:00	22.09.23 14:00	207.052	213.005	212.966	67.8678	NA	NA	NA
U10	10	14.09.23 21:00	24.09.23 14:00	205.658	170.74	169.976	38.3169	45.909	45.909	3.5346
C10	10	14.09.23 21:00	24.09.23 14:00	191.372	193.595	190.069	45.4224	NA	NA	NA
U11	11	14.09.23 21:00	24.09.23 14:00	207.417	170.888	169.844	48.4498	61.839	61.839	4.515
C11	11	14.09.23 21:00	24.09.23 14:00	206.489	208.773	206.423	56.7002	NA	NA	NA
U12	12	14.09.23 21:00	24.09.23 14:00	201.093	168.277	161.599	46.2318	51.8	51.8	4.0106
C12	12	14.09.23 21:00	24.09.23 14:00	204.401	205.934	204.571	62.458	NA	NA	NA
U13	13	14.09.23 21:00	26.09.23 15:00	198.336	179.044	177.221	50.1837	23.332	22.315	1.5505
C13	13	14.09.23 21:00	26.09.23 15:00	201.081	206.309	204.088	59.5323	NA	NA	NA
U14	14	14.09.23 21:00	26.09.23 15:00	203.372	169.069	166.856	48.3916	40.678	39.651	3.2069
C14	14	14.09.23 21:00	26.09.23 15:00	201.362	206.423	204.183	58.2674	NA	NA	NA
U15	15	14.09.23 21:00	26.09.23 15:00	200.244	170.391	168.895	43.0646	39.804	38.732	3.171
C15	15	14.09.23 21:00	26.09.23 15:00	200.547	210.462	208.532	49.3328	NA	NA	NA

Tank = tank identity where prefix U means urchin treatment and prefix C means autogenic control, **Sporophyte** = sporophyte identity, **Deployment** = deployment date and time as dd.mm.yy hh:mm, **Retrieval** = retrieval date and time as dd.mm.yy hh:mm, **Initial** = initial lamina dripped wet or buoyant mass, **Final** = final lamina dripped wet or buoyant mass, **Lamina.wet** = remaining lamina dripped wet or buoyant mass after excision of disc for mesograzers experiment, **Lamina.dry** = remaining lamina mass after freeze-drying, **Faeces** = buoyant mass of produced faeces, **Faeces.wet** = remaining faecal buoyant mass after sampling for mesograzers experiment, **Faeces.dry** = remaining faecal mass after freeze-drying, $Lamina.dry:Lamina.wet$ and $Faeces.dry:Faeces.wet$ represent dry-wet mass ratios, all masses are given in g

Urchin mass and size change

Tank	Deployment	Retrieval	Initial	Final	Diameter
U1	14.09.23 21:00	18.09.23 16:00	13.614	13.961	33.71
U1	14.09.23 21:00	18.09.23 16:00	44.029	44.919	51.14
U1	14.09.23 21:00	18.09.23 16:00	38.141	38.941	47.68
U1	14.09.23 21:00	18.09.23 16:00	36.141	36.489	47
U1	14.09.23 21:00	18.09.23 16:00	50.242	50.474	51.94
U2	14.09.23 21:00	18.09.23 16:00	62.315	63.175	56.01
U2	14.09.23 21:00	18.09.23 16:00	31.244	31.97	41.41
U2	14.09.23 21:00	18.09.23 16:00	13.975	13.871	32.66
U2	14.09.23 21:00	18.09.23 16:00	32.702	33.651	44.08
U2	14.09.23 21:00	18.09.23 16:00	26.485	26.775	43.53
U3	14.09.23 21:00	18.09.23 16:00	41.585	43.486	49.78
U3	14.09.23 21:00	18.09.23 16:00	32.613	33.034	44.8
U3	14.09.23 21:00	18.09.23 16:00	39.018	39.788	46.12
U3	14.09.23 21:00	18.09.23 16:00	51.02	51.984	48.47
U3	14.09.23 21:00	18.09.23 16:00	18.097	18.592	35.93
U4	14.09.23 21:00	20.09.23 15:00	21.225	21.687	36.67
U4	14.09.23 21:00	20.09.23 15:00	42.018	42.542	49.8
U4	14.09.23 21:00	20.09.23 15:00	47.147	47.91	52
U4	14.09.23 21:00	20.09.23 15:00	24.935	26.008	39.43
U4	14.09.23 21:00	20.09.23 15:00	20.464	20.832	36.2
U5	14.09.23 21:00	20.09.23 15:00	26.538	27.593	40.07
U5	14.09.23 21:00	20.09.23 15:00	16.522	17.242	35.67
U5	14.09.23 21:00	20.09.23 15:00	29.666	30.674	41.15
U5	14.09.23 21:00	20.09.23 15:00	36.3	37.679	44.47
U5	14.09.23 21:00	20.09.23 15:00	24.591	24.81	38.68
U6	14.09.23 21:00	20.09.23 15:00	59.213	60.562	54.64
U6	14.09.23 21:00	20.09.23 15:00	49.694	50.507	51.26
U6	14.09.23 21:00	20.09.23 15:00	24.067	25.051	39.48
U6	14.09.23 21:00	20.09.23 15:00	34.381	35.622	43.98
U6	14.09.23 21:00	20.09.23 15:00	40.566	42.166	46.39
U7	14.09.23 21:00	22.09.23 14:00	39.404	39.313	46
U7	14.09.23 21:00	22.09.23 14:00	31.718	32.289	43.69
U7	14.09.23 21:00	22.09.23 14:00	62.762	64.281	55.89
U7	14.09.23 21:00	22.09.23 14:00	24.136	24.936	38.47
U7	14.09.23 21:00	22.09.23 14:00	49.582	50.55	50.98
U8	14.09.23 21:00	22.09.23 14:00	33.448	33.961	44.01
U8	14.09.23 21:00	22.09.23 14:00	43.404	44.26	48.13
U8	14.09.23 21:00	22.09.23 14:00	32.138	33.118	42.06
U8	14.09.23 21:00	22.09.23 14:00	34.067	35.175	45.94
U8	14.09.23 21:00	22.09.23 14:00	41.569	42.119	46.4
U9	14.09.23 21:00	22.09.23 14:00	60.972	60.967	53.43
U9	14.09.23 21:00	22.09.23 14:00	73.423	74.33	57.79
U9	14.09.23 21:00	22.09.23 14:00	29.547	30.125	42.62
U9	14.09.23 21:00	22.09.23 14:00	37.022	38.176	45.89
U9	14.09.23 21:00	22.09.23 14:00	28.77	30.079	40.53
U10	14.09.23 21:00	24.09.23 14:00	46.983	48.3	49.72
U10	14.09.23 21:00	24.09.23 14:00	69.146	68.752	55.9
U10	14.09.23 21:00	24.09.23 14:00	28.035	28.21	39.75
U10	14.09.23 21:00	24.09.23 14:00	36.089	36.292	43.06
U10	14.09.23 21:00	24.09.23 14:00	38.386	38.65	46.23
U11	14.09.23 21:00	24.09.23 14:00	49.511	49.853	50.85

U11	14.09.23 21:00	24.09.23 14:00	36.152	36.533	44.47
U11	14.09.23 21:00	24.09.23 14:00	67.192	67.74	54.71
U11	14.09.23 21:00	24.09.23 14:00	40.97	41.1	46.99
U11	14.09.23 21:00	24.09.23 14:00	40.797	40.612	46.96
U12	14.09.23 21:00	24.09.23 14:00	54.795	55.569	50.86
U12	14.09.23 21:00	24.09.23 14:00	29.635	31.22	41.55
U12	14.09.23 21:00	24.09.23 14:00	41.577	42.246	46.83
U12	14.09.23 21:00	24.09.23 14:00	45.579	45.91	48.93
U12	14.09.23 21:00	24.09.23 14:00	36.887	37.791	44.28
U13	14.09.23 21:00	26.09.23 15:00	58.631	58.18	52.48
U13	14.09.23 21:00	26.09.23 15:00	36.073	37.146	45.4
U13	14.09.23 21:00	26.09.23 15:00	15.061	15.117	32
U13	14.09.23 21:00	26.09.23 15:00	27.637	27.792	40.31
U13	14.09.23 21:00	26.09.23 15:00	34.548	35.235	43.31
U14	14.09.23 21:00	26.09.23 15:00	39.645	38.798	46.77
U14	14.09.23 21:00	26.09.23 15:00	70.291	70.854	55.42
U14	14.09.23 21:00	26.09.23 15:00	43.483	44.109	47.1
U14	14.09.23 21:00	26.09.23 15:00	21.392	22.127	37.46
U14	14.09.23 21:00	26.09.23 15:00	22.038	22.783	38.1
U15	14.09.23 21:00	26.09.23 15:00	70.115	70.483	56.83
U15	14.09.23 21:00	26.09.23 15:00	33.158	33.463	41.39
U15	14.09.23 21:00	26.09.23 15:00	36.707	37.67	45.54
U15	14.09.23 21:00	26.09.23 15:00	20.206	21.563	37.27
U15	14.09.23 21:00	26.09.23 15:00	19.707	20.565	36.45

Tank = tank identity, **Deployment** = deployment date and time as dd.mm.yy hh:mm, **Retrieval** = retrieval date and time as dd.mm.yy hh:mm, **Initial** = initial urchin dripped wet mass, **Final** = final urchin dripped wet mass, **Diameter** = urchin test diameter (I only measured test diameter once, assuming that it would not change), masses are given in g and diameter in mm

Suggested mesograzers variables

Tank = urchin tank from which material was sourced, **Sporophyte** = sporophyte identity, **Arena** = grazing arena identity where prefix G means grazer treatment and prefix C means autogenic control, **Deployment** = deployment date and time as dd.mm.yy hh:mm, **Retrieval** = retrieval date and time as dd.mm.yy hh:mm, **Grazer.initial** = initial mesograzers buoyant mass, **Grazer.final** = final grazer buoyant mass, **Lamina.initial** = initial lamina disc buoyant mass, **Lamina.final** = final lamina buoyant mass, **Lamina.final.dry** = final lamina mass after freeze-drying, **Faeces.initial** = initial buoyant mass of urchin faeces, **Faeces.final.dry** = final mass of urchin faeces after freeze-drying, all masses are given in g

Relevant papers

- Anderson RJ, Velimirov B. 1982.** An experimental investigation of the palatability of kelp bed algae to the sea urchin *Parechinus angulosus* Leske. *Marine Ecology* **3**: 357–373.
- Dethier MN, Hoins G, Kobelt J et al. 2019.** Feces as food: the nutritional value of urchin feces and implications for benthic food webs. *J Exp Mar Biol Ecol* **514-515**: 95–102.
- Filbee-Dexter K, Scheibling RE. 2014a.** Sea urchin barrens as alternative stable states of collapsed kelp ecosystems. *Marine Ecology Progress Series* **495**: 1–25.
- Filbee-Dexter K, Scheibling RE. 2014b.** Detrital kelp subsidy supports high reproductive condition of deep-living sea urchins in a sedimentary basin. *Aquatic Biology* **23**: 71–86.
- Filbee-Dexter K, Pedersen MF, Fredriksen S et al. 2020.** Carbon export is facilitated by sea urchins transforming kelp detritus. *Oecologia* **192**: 213–225.
- Gilson AR, Smale DA, O'Connor N. 2021.** Ocean warming and species range shifts affect rates of ecosystem functioning by altering consumer-resource interactions. *Ecology* **102**: e03341.
- Hargrave MS, Foggo A, Pessarrodona A, Smale DA. 2017.** The effects of warming on the ecophysiology of two co-existing kelp species with contrasting distributions. *Oecologia* **183**: 531–543.
- Johnson CR, Mann KH. 1986.** The importance of plant defence abilities to the structure of subtidal seaweed communities: the kelp *Laminaria longicruris* de la Pylaie survives grazing by the snail *Lacuna vineta* (Montagu) at high population densities. *Journal of Experimental Marine Biology and Ecology* **97**: 231–267.
- Lemire M, Himmelman JH. 1996.** Relation of food preference to fitness for the green sea urchin, *Strongylocentrotus droebachiensis*. *Marine Biology* **127**: 73–78.
- Mamelona J, Pelletier É. 2005.** Green urchin as a significant source of fecal particulate organic matter within nearshore benthic ecosystems. *Journal of Experimental Marine Biology and Ecology* **314**: 163–174.
- Molis M, Körner J, Ko YW, Kim JH. 2008.** Specificity of inducible seaweed anti-herbivory defences depends on identity of macroalgae and herbivores. *Marine Ecology Progress Series* **354**: 97–105.
- Molis M, Enge A, Karsten U. 2010.** Grazing impact of, and indirect interactions between mesograzers associated with kelp (*Laminaria digitata*). *Journal of Phycology* **46**: 76–84.
- Ng CA, Micheli F. 2020.** Size-dependent vulnerability to herbivory in a coastal foundation species. *Oecologia* **193**: 199–209.

- Norderhaug KM, Fredriksen S, Nygaard K. 2003.** Trophic importance of *Laminaria hyperborea* to kelp forest consumers and the importance of bacterial degradation to food quality. *Mar Ecol Prog Ser* **255**: 135–144.
- Pavia H, Toth GB. 2000.** Inducible chemical resistance to herbivory in the brown seaweed *Ascophyllum nodosum*. *Ecology* **81**: 3212–3225.
- Pessarrodona A, Foggo A, Smale DA. 2019.** Can ecosystem functioning be maintained despite climate-driven shifts in species composition? Insights from novel marine forests. *J Ecol* **107**: 91–104.
- Poore AG, Gutow L, Pantoja JF, Tala F, Jofré Madariaga D, Thiel M. 2014.** Major consequences of minor damage: impacts of small grazers on fast-growing kelps. *Oecologia* **174**: 789–801.
- Robertson AI, Lucas JS. 1983.** Food choice, feeding rates, and the turnover of macrophyte biomass by a surf-zone inhabiting amphipod. *Journal of Experimental Marine Biology and Ecology* **72**: 99–124.
- Rohde S, Molis M, Wahl M. 2004.** Regulation of anti-herbivore defence by *Fucus vesiculosus* in response to various cues. *Journal of Ecology* 1011–1018.
- Ruocco N, Zupo V, Caramiello D et al. 2018.** Experimental evaluation of the feeding rate, growth and fertility of the sea urchins *Paracentrotus lividus*. *Invertebrate Reproduction & Development* **62**: 209–220.
- Sauchyn LK, Scheibling RE. 2009a.** Fecal production by sea urchins in native and invaded algal beds. *Marine Ecology Progress Series* **396**: 35–48.
- Sauchyn LK, Scheibling RE. 2009b.** Degradation of sea urchin feces in a rocky subtidal ecosystem: implications for nutrient cycling and energy flow. *Aquat Biol* **6**: 99–108.
- Scheibling RE, Hatcher BG. 2007.** Ecology of *Strongylocentrotus droebachiensis*. *Developments in Aquaculture and Fisheries Science* 37. Elsevier, 353–392.
- Vasseur E. 1952.** Geographic variation in the Norwegian sea-urchins, *Strongylocentrotus droebachiensis* and *S. pallidus*. *Evolution* 87–100.
- Vergés A, Alcoverro T, Romero J. 2011.** Plant defences and the role of epibiosis in mediating within-plant feeding choices of seagrass consumers. *Oecologia* **166**: 381–390.
- Wernberg T, Filbee-Dexter K. 2018.** Grazers extend blue carbon transfer by slowing sinking speeds of kelp detritus. *Sci Rep* **8**: 17180.
- Wessels H, Hagen W, Molis M, Wiencke C, Karsten U. 2006.** Intra- and interspecific differences in palatability of Arctic macroalgae from Kongsfjorden (Spitsbergen) for two benthic sympatric invertebrates. *Journal of experimental marine biology and ecology* **329**: 20–33.
- Yorke CE, Page HM, Miller RJ. 2019.** Sea urchins mediate the availability of kelp detritus to benthic consumers. *Proc R Soc B* **286**: 20190846.